HACK4HUMANITY 2026 · ROUND 1 SOLUTION DOCUMENT

Track C — AI for Societal Good | IEEE JCTS · IEEE SIGHT · BRAIN Foundation, Pune

KisaanRaksha

किसानरक्षा

Farmer Financial Distress Early Warning System — Maharashtra

An AI agent that predicts farmer financial crisis before it happens, alerts officers proactively, and helps farmers file insurance claims in Marathi over WhatsApp voice.

SDG 2 · Zero Hunger **SDG 3 · Good Health & Well-Being** **SDG 10 · Reduced Inequalities**

Team: Sahil Karande · Pranav Shripannavar

GitHub: github.com/sahilkarande0918-cmd/Kisan-Raksha-project

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1. The Problem — A Farmer Dies Every 3 Hours

Maharashtra loses a farmer to suicide every 3 hours. In 2024, **2,706 farmers died in Vidarbha and Marathwada alone** — only 1,563 were even deemed eligible for aid. State response is entirely reactive: ex-gratia compensation paid after a death, with zero predictive infrastructure.

Yet the crisis follows a measurable pattern that builds over months:

- A failed monsoon (rainfall deficit) destroys the kharif crop;
- mandi prices fall below MSP just when the farmer must sell;
- the crop-loan repayment deadline arrives with nothing to pay it.

Each signal is publicly observable in real time — rainfall archives, Agmarknet prices, satellite crop health, the crop calendar. **The data to see the crisis coming already exists. Nobody is looking.**

North-star beneficiary: *a 2-acre cotton farmer in Amravati with a ₹1.2 lakh crop loan, a WhatsApp feature phone, and Marathi as his only fluent language. Every design choice traces back to him — voice-first, Marathi-first, no new app, offline-tolerant.*

2. The Solution — Predict · Alert · Assist

- **PREDICT —** a LightGBM model fuses four live signals into a Financial Stress Index (FSI 0–100) per district; FSI > 75 = critical.
- **ALERT —** on a critical FSI the duty officer gets a WhatsApp alert with the evidence — deficit %, price gap, NDVI — before a crisis, not after a death.
- **ASSIST —** the farmer sends a Marathi voice note (or text) on WhatsApp — no app, no typing, no English — and receives a grounded reply plus an auto-drafted PMFBY insurance-claim letter.

16 districts covered (Vidarbha + Marathwada)	**4** grounded live signals fused per district	**87.9** FSI in labeled crisis simulation → alert + claim fired on a real phone

3. Architecture — Custom MCP Server, No Bare LLM Calls

Every fact the agent states comes from a role-scoped MCP tool grounded in a real dataset — the LLM never answers from training data (track slide 7).

Farmer WhatsApp Marathi voice	→	**Agent / LLM** tool-calling loop (Groq / Gemini)	→	**MCP Server** FastMCP — 7 role-scoped tools	→	**Grounded Data** Open-Meteo · Agmarknet · NASA MODIS

Inbound: Twilio WhatsApp → FastAPI webhook → Whisper large-v3 ASR (Marathi/Hindi auto-detect) → agent. Outbound: Twilio for farmer replies, officer alerts and claim PDFs. A Streamlit choropleth gives officers the live FSI heatmap.

3.1 The seven MCP tools

Tool	One clear job	Grounding
query_weather_signal	30-day rainfall deficit vs 5-yr baseline	Open-Meteo ERA5
query_mandi_prices	median mandi price vs MSP	Agmarknet (data.gov.in)
query_ndvi_signal	satellite crop-health stress	NASA MODIS MOD13Q1
compute_fsi	fuse 4 signals via LightGBM → FSI 0–100	trained model, held-out
get_farmer_history	session memory (hashed phone key)	SQLite — no raw PII
send_officer_alert	WhatsApp alert with evidence	Twilio
draft_pmfby_claim	Marathi PMFBY letter from evidence	LLM + FSI evidence

One job per tool (slide 9); every decision traceable through logged calls; tools are auditable and reusable as-is by another team.

4. Technical Methodology

4.1 Four signals → Financial Stress Index

Signal	Source (live API)	What it measures
Drought probability	Open-Meteo ERA5 (keyless)	30-day rainfall vs 5-yr average
Mandi price deviation	Agmarknet, data.gov.in	% of median modal price below MSP
NDVI satellite index	NASA MODIS MOD13Q1	crop canopy health from space
Repayment proximity	district crop calendar	nearness to loan-due window (amplifier)

Drought **Price gap vs MSP** **NDVI (satellite)** **Repayment proximity**	→	**LightGBM** gradient-boosted fusion	→	**FSI 0–100** > 75 = CRITICAL alert

4.2 The FSI model — what its metrics do and do not mean

A LightGBM regressor maps the four signals plus month/crop/region to FSI 0–100, trained on our 13,644-row synthetic dataset (§6) and evaluated on **held-out talukas the model has never seen**.

Consistency check on synthetic data (held-out talukas)	Value
Mean absolute error (0–100 scale)	3.75
R^2	0.942
Critical-alert recall (FSI > 75)	0.78
Critical-alert precision	0.83

What these numbers mean, plainly: our synthetic labels come from a domain-logic formula, so the model is recovering a function we designed. The metrics measure **internal pipeline consistency —**

not real-world predictive power. Real validation is the §8 pilot: alert precision vs officer-verified ground truth over one season. Scoping the ML claim honestly is deliberate.

Threshold FSI > 75 is a tunable operating point (recall 0.78 / precision 0.83 ≈ one false alert per four), calibrated per district against officer capacity in the pilot. The label's interaction terms (drought × repayment, price × repayment) encode the core insight: distress spikes when crop failure meets the loan deadline — the documented Oct–Jan clustering.

4.3 Context engineering (track slide 8 — all six elements)

Track requirement	KisaanRaksha implementation
Tool-grounded reasoning	agent calls MCP tools for every fact; zero bare answers
Structured memory	SQLite history: district, crop, past FSI, claims — recalled each turn
Localization-aware	Marathi-first; Hindi/English auto-matched; ₹/quintal; Devanagari spellings pinned
Retrieval grounding	crop calendar, MSP tables, district registry as curated context
Guardrails	FSI thresholds decide alerts (not the LLM); SIMULATION always labeled; helpline in critical replies
Multi-step planning	plan → signals → FSI → alert → claim, state carried across steps

5. Responsible AI — §8.3 Compliance & Bias Audit

The FSI directs officer attention, so we audit the critical-alert decision (FSI > 75) on held-out talukas. The gated harm metric is the false-negative rate — a missed alert is a missed farmer.

Group	Alert rate	False-negative rate	MAE
Vidarbha	0.064	0.205	3.74
Marathwada	0.077	0.229	3.75
Gap across regions	0.013	0.024	0.015
Gap across crops	0.045	0.074	0.23

Fairness gate	Threshold	Observed	Result
FNR gap — region	< 0.10	0.024	PASS
FNR gap — crop	< 0.10	0.074	PASS
MAE gap — region	< 2.0 pts	0.015	PASS
MAE gap — crop	< 2.0 pts	0.23	PASS
Alert-rate gap	monitored, not gated	0.013 / 0.045	—

FNR gaps are gated at 0.10, MAE gaps at 2.0 points. Alert-rate is monitored not gated — Marathwada's higher rate reflects genuinely higher drought risk; equalizing it would suppress true alerts. Full report: model/fairlearn_report.md.

5.2 Data ethics & honest limitations

- No real farmer PII: training data is synthetic; live farmers keyed by hashed WhatsApp numbers.
- Every source cited (README): Open-Meteo CC-BY 4.0, Agmarknet, NASA MODIS, CACP MSP, public GeoJSON.
- The demo crisis is always labeled 'SIMULATION' — no fabricated live claims.
- Metrics show pipeline consistency on synthetic data, not real-world prediction (see §4.2, §8 pilot).
- District-level signals mask farm variance; the FSI is an officer-attention tool, never an automated eligibility or denial decision.

6. Custom Dataset — Flagged for IEEE Dataport

maharashtra_ag_stress_dataset.csv — 13,644 (district × taluka × crop × year × month) rows for 16 districts, with the four signals and a documented stress label. Fully synthetic (zero PII), reproducible from a seeded script, and grounded in real distributions: IMD drought propensity, 2024–26 Agmarknet-vs-MSP gaps, MODIS kharif NDVI seasonality, real CACP MSP. Flagged for IEEE Dataport (slide 11); methodology in dataset/README.md.

7. Impact — Accessibility & Scalability

- **Accessibility:** WhatsApp voice in the farmer's own language — no app, no literacy requirement, any phone; offline-first server with cached fallback for low connectivity.
- **Scalability:** all sources are pan-India public APIs; adding a district is one JSON entry; free-tier inference means fractions of a rupee per conversation; the four-signal pattern generalizes to any rain-fed belt.

Deployment metric	Current build	State scale-up path
Districts monitored	16 (Vidarbha + Marathwada)	36 (all Maharashtra) — config only
Farmers reachable	every WhatsApp user in coverage	WhatsApp Business API + Krishi Vibhag lists
Claim assistance	PMFBY letter auto-drafted (Marathi)	direct PMFBY portal API
Officer surface	WhatsApp alerts + Streamlit heatmap	integrate into Mahavedh / Krishi dashboards

8. Deployment Pathway & Sustainability

- Ownership: handoff to Krishi Vibhag (district offices) or an NGO — the alert flow mirrors their duty-roster practice.
- Low cost: one <1 MB LightGBM file, free-tier inference, no GPU in the serving path.
- Maintainable: modular repo, one-command dataset regen + retrain, bias audit regenerates automatically.
- Pilot: one monsoon season across 3 Amravati talukas, measuring alert precision vs officer-verified ground truth and PMFBY completion rates.

9. Live Demo — Verified End-to-End on a Real Phone

Executed on a physical phone via the Twilio WhatsApp sandbox. The farmer can send a **Marathi voice note or a text message** — voice is the primary, literacy-free path; Whisper transcribes it either way. The agent then chains weather → mandi → NDVI → compute_fsi and branches on the score:

Marathi voice WhatsApp note	→	**Whisper ASR** → text	→	**Agent + MCP** 4 signals → FSI	→	**FSI 0–100** threshold 75

FSI < 75 → LOW honest all-clear reply, no alert		**FSI > 75 → CRITICAL** officer alert + Marathi PMFBY claim PDF

Crisis path (labeled simulation).

A demo message replaying the Oct–Nov 2024 stress pattern raises FSI to 87.9 CRITICAL. The officer receives a WhatsApp alert with the full evidence; the farmer receives Marathi guidance and an auto-drafted PMFBY claim as a PDF. Every simulated message carries a visible SIMULATION tag.

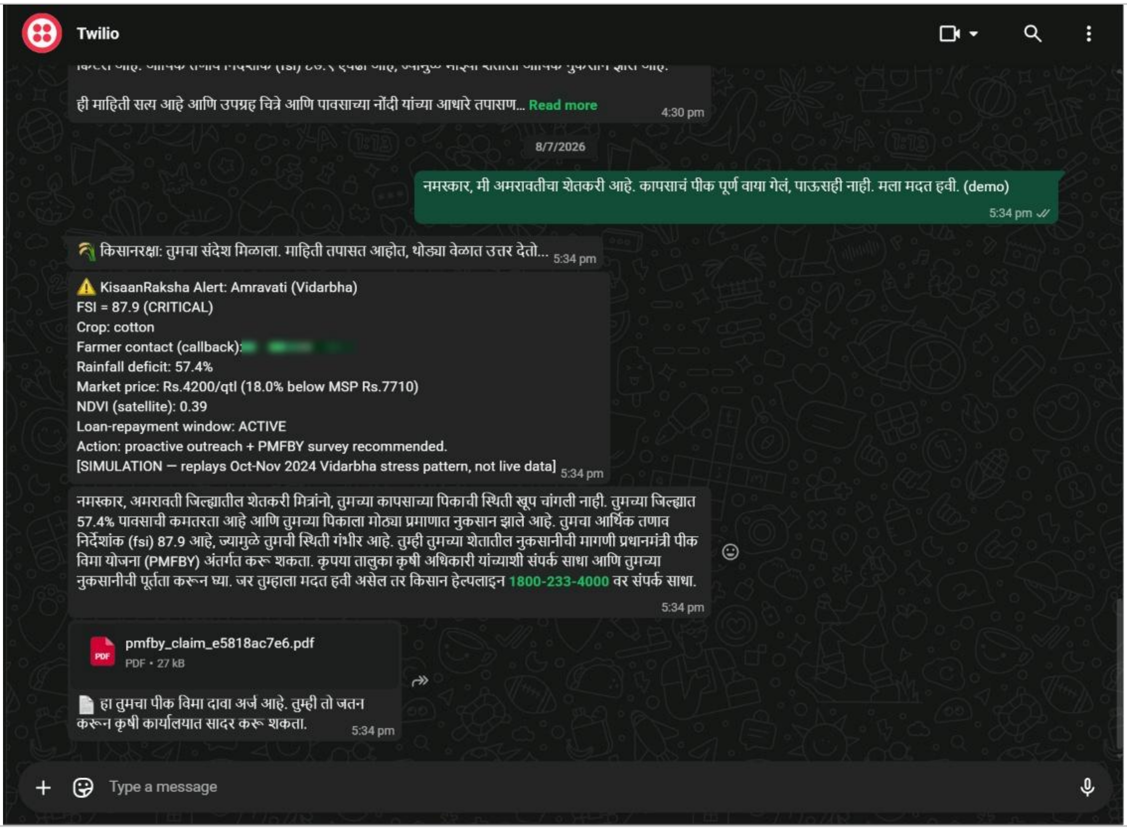

Fig. 1 — Crisis simulation: officer alert (FSI 87.9 CRITICAL, evidence + farmer callback), Marathi guidance with the Kisan helpline, and the auto-drafted PMFBY claim delivered as a PDF.

Calm path (live data).

On a real, calm day the same pipeline returns FSI 20.9 LOW and replies with an honest all-clear — no alert, no claim. The contrast between the two is the point: the system does not cry wolf.

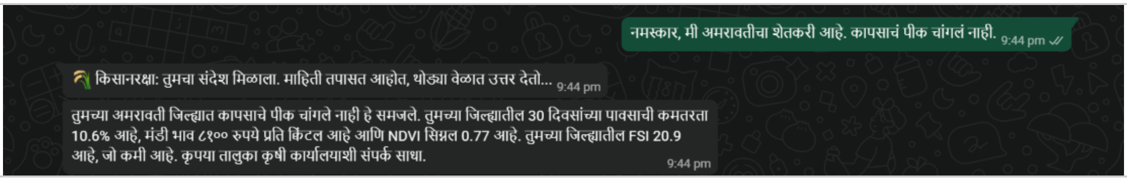

Fig. 2 — Calm day on live data: rainfall deficit 10.6%, NDVI 0.77, FSI 20.9 LOW → all-clear Marathi reply, no false alarm.

10. Team, Repository & Declaration

Item	Detail
Team	Sahil Karande · Pranav Shripannavar — 2 members
Repository	github.com/sahilkarande0918-cmd/Kisan-Raksha-project
Stack	Python · FastMCP · FastAPI · LightGBM · Fairlearn · Whisper/AI4Bharat ASR · Twilio WhatsApp · Streamlit · SQLite
Custom dataset	dataset/maharashtra_ag_stress_dataset.csv — flagged for IEEE Dataport
Bias audit	model/fairlearn_report.md — all gates pass
Declaration	All data public or synthetic; no real personal data collected; per Code of Conduct §8.3.

Technology must serve humanity — KisaanRaksha points it at the farmer the system forgot.